

TYPED RETURN MAPS THROUGH THE ES-FABLE-LEECH PROGRAMME COMPOSITION AUDIT, EXACT REPAIRS, AND SOURCE-ADDRESSABLE PROOF MODULES

THE CLANKERS: WORKBENCH TRANCHE

ABSTRACT. We audit eight source packages at their typed interfaces, preserving selector markings, all four Cayley faces, factor order, coefficient fields, lattice models, return maps and finite scopes. New composition calculations exhibit a nine-trace symmetry obstruction, provide a full-octad inverse with an exact mod-four syndrome, and construct an octad-aligned Golay–Wilson isometry. The full Albert cubic correction is retained. Two output-validation gaps are reproduced and repaired. Three short proof modules isolate the resulting mathematical claims; a 49-arrow register states all domains, fibres and reconstruction data. No conjecture resolution, independent review or formal-prover certification is asserted.

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1. SCOPE, EVIDENCE, AND THE RULE FOR AN INVERSE CLAIM

This audit uses eight locally available mathematical packages, beginning with the marked exterior/middle selector and ending with the three-octad Leech inverse. The exact 105 source files and their SHA-256 identities are in `source_inventory.json`. It does not reconstruct an unavailable conversation or certify an uninspected Git branch. The original files are preserved unchanged under `inputs/`.

Eighteen original execution processes returned zero exit status. The first two reproduced the original bounded arithmetic census and the preserved covering-certificate replay through 10^{10} . The other eight checker families were each run normally and under Python optimization, with byte-identical paired output files. The first two assertion-using programs intentionally reject optimized execution. Re-running a program is not an independent reviewer or a proof of every written assertion. The new checker is separately written, uses only Python’s standard library, and exercises the interfaces that the original per-package tests did not join.

A many-to-one morphism is legitimate. What requires proof is the inference through it: either retain its complete fibre, or prove that the property being transported is constant on that fibre. Recording a discarded coordinate does not by itself prove an inverse; the return equations below are checked on their specified domains.

1.1. The three substantive interface repairs. The earlier nine-coordinate time quotient is valid for the cyclic time operator T , but not for the second generator J . Its smallest linear completion for both actions is

the full 24-dimensional coordinate space, represented here by 759 octad traces. Their real image and single modulo-four integral obstruction are explicit.

The projective Golay and Wilson octonionic Leech models previously occurred without a chosen intervening isometry. An octad-aligned signed permutation now supplies that arrow, with equality of infinite lattices proved using 37 generators and unimodularity. Wilson already gives an isometry to MOG coordinates; no first-existence claim is attached to this convention-compatible choice.

The cubic norm of the octonionic embedding is not determined by the three traces and is not invariant under every Leech coordinate isometry. Explicit vectors show both failures. Keeping the entire lattice coordinate, or its complete trace packet, retains the missing data. The corrected Albert–ES polynomial relation includes all mixed terms.

2. THE COMPLETE MARKED ARITHMETIC–FACTOR JUNCTION

Write a selector state as

$$m = (p, R, u_E, h, r, s, c, q; x, y, z), \quad c \in \{E, M\}.$$

The subscript on u_E distinguishes the ES divisor from the Fable source coordinate u_F ; it does not restrict the channel to E . All denominators are ordered. On the inherited bounded domain, p is odd prime, $0 < R < p$, $p + R = 4hrs$, $u_E = hr^2$ and $\gcd(r, s) = 1$. The channel formulas are

c	q	(x, y, z)
E	$(pr + s)/R$	$(hrs, hsq, phrq)$
M	$(r + s)/R$	$(hrs, phsq, phrq)$.

The quotients must be positive integers. The inverse divisor formulas are

$$u_E = \frac{x^2}{Ry - px} \quad (E), \quad u_E = \frac{px^2}{Ry - px} \quad (M).$$

With $d = \gcd(x, u_E)$, the inverse is $r = u_E/d$, $s = x/d$, $h = d/r$. These are canonical factor-exponent equalities, not choices from an unbounded generated subgroup. The original Yamamoto/Elsholtz–Tao lineage and exact references are retained in the source package; the mapped identities here can also be checked by clearing their displayed denominators.

2.1. Four faces, three distinguished factors, and two orientations. Set $a = (-p, 4x, 4y, 4z)$ and retain it affinely. It satisfies $e_3(a) = 0$, equivalently $\sum_i a_i^{-1} = 0$. For a flag $(i, j, k, l) \in S_4$, put

$$h_i = -\frac{1}{4a_i}, \quad (r_0, r_1, r_2) = \left(-\frac{4a_i}{a_j}, -\frac{4a_i}{a_k}, -\frac{4a_i}{a_l} \right).$$

Their sum is four. The complete open domain here requires nonzero, pairwise distinct a_i ; hence the roots are nonzero, pairwise distinct and not -4 . The inverse is

$$a_i = -\frac{1}{4h_i}, \quad (a_j, a_k, a_l) = \frac{1}{h_i}(r_0^{-1}, r_1^{-1}, r_2^{-1}).$$

The affine scale h_i is essential for recovering p and the denominator marking.

The normalized cubic is

$$f(Z) = -\frac{1}{4}(Z - r_0)(Z - r_1)(Z - r_2) = -\frac{1}{4}Z^3 + Z^2 + \frac{\beta}{2}Z + \alpha.$$

Retain $r = r_0$, $d = f'(r) \neq 0$ and $\eta = r_1 - r_2$. Then

$$\eta^2 = (3r - 4)^2 + 16d, \quad (r_1, r_2) = ((4 - r + \eta)/2, (4 - r - \eta)/2).$$

The derivative and its square-root discriminant are insufficient without the sign of η when companion order is required. The oriented face change is

$$(r_0, r_1, r_2) \mapsto (16/r_0, -4r_2/r_0, -4r_1/r_0), \\ (r, d, \eta) \mapsto (16/r, 8 + 16(d - 8)/r^2, 4\eta/r).$$

It exchanges the two leading flag indices and the two companion indices, preserving parity. These formulas are rational involutions on the stated domain.

The exact Fable source point for a finite selected root is

$$(u_F, v_F, w_F) = (d^{-1}, -r - d, 5d^2 + 3rd + \frac{1}{2}d^3).$$

The inverse selected-root data are $d = 1/u_F$ and $r = -v_F - d$. In the general target cubic, $\frac{1}{2}d^3$ is replaced by $-\gamma d^3$. The infinity branch is separately present when $\gamma = 0$; it must not be obtained by substituting infinity into the finite formula. The underlying normalized factor construction is Tao's equations (2), (5), (7), and (10).

On the distinct-root locus, forgetting the companion order is $2 : 1$, forgetting the selected root is $3 : 1$, and forgetting all root order is $6 : 1$. Four face choices yield 12 orientation-preserving flags or 24 flags with both orientations. The 12 fixed-cubic tetrahedral pencil states in the earlier note are a different fourfold fibre over the three selected factors. They are not twelve geometric inverse points of one unmarked polynomial map.

2.2. One complete end-to-end test. For each of the states

p	c	R	(x, y, z)
5	E	3	(2, 4, 20)
1201	E	23	(306, 16218, 1082101)
2521	M	31	(638, 804199, 55462)

the new checker traverses all 24 flags, forms all three local normalized factors, checks the resultant and affine normalization, returns from the selected Fable point using η and h_i , reconstructs the original ordered denominators, lifts them to Λ , transports through Wilson and compact-Zorn/first-Tits coordinates, returns to the lattice and its traces, and finally reconstructs every selector field. All 72 full returns pass. There are 216 distinct face/local-factor instances and 432 factor-return evaluations including the face-exchanged instances. These are fixed-instance checks; the general inverse claims use the rational and linear identities just displayed.

3. A FULL STATE, RATHER THAN AN UNPROVED IDENTIFICATION

For a marked arithmetic state m let $d(m)$ be its ordered denominator triple. Define

$$\mathcal{R} = \{(m, \lambda) : m \text{ is an actual selector state, } P\lambda = d(m)\}.$$

If $K = \ker P$ and σ is the integral section, then

$$(m, k) \mapsto (m, \sigma(d(m)) + k), \quad (m, \lambda) \mapsto (m, \lambda - \sigma(d(m)))$$

are exact inverse maps. Attach the chosen flag, factor data, and scale by the preceding formulas. Attach Wilson and Albert coordinates by the explicit isometry in the next note. This forms a commutative reconstruction diagram, not an assertion that a finite A_4 label encodes p . The finite flag-to-projective-coordinate table carries the labels; the arithmetic state remains a separate field.

The distinction between the original full-shell system $R < 2p$ and the bounded selector $R < p$ remains in the type. Chamber positivity, when required, is an extra exact inequality; an unmarked positive denominator solution is not automatically an image of the chosen chamber-restricted family. Neither a change of type nor a finite verification range silently enlarges these domains.

4. FIELD AND ANALYTIC INTERFACES THAT REMAIN SEPARATE

The three-slot associator of the supplied algebra is vector-valued. Its 27 input cells are vectors, hence 81 scalar coefficients when the output is three-dimensional. One scalar contraction loses data; the entire dual family reconstructs the tensor. Likewise, three compatible marked bilinear faces of a scalar cube leave an eight-dimensional affine fibre. The common determinant equation of three cube slices does not specify their incidence maps or retained return translation.

The cube-to- D_T construction is injective as a linear map of 27 tensor coefficients into operators. Its nilpotence identity $D_u^2 = 0$, and thus the formula $I + tD_u$, were proved for pure $u = a \otimes b \otimes c$. They are not automatically valid for an arbitrary sum of pure tensors.

The first-Tits presentation is split over its stated base field. Compact octonions are recovered only after retaining the real structure $A = A^*$, $C = B^*$. For example, start with $A = \text{diag}(2, 3, 5)$ and $B = C = 0$. The pure tensor (e_1, e_2, e_3) at parameter one produces $B = -2E_{33}$, $C = 0$. Its cubic norm is still 30, but $C \neq B^*$; the compact inverse correctly rejects it. This is a domain test, not a refutation of the split norm identity.

Reduction of rational projective marks modulo 23 is a specialization with many arithmetic lifts, not an injective map from complex values. The three named half-marks and all finite branch labels may be carried, but a complex Mellin argument is not reduced modulo 23. Similarly, the 24 named classes inject into $\Lambda/12\Lambda$, whereas the quotient of the full lattice has kernel 12Λ . The full finite Fourier return uses 12^{24} classes and is not closed on only the 24 named classes.

The Mellin normalization remains $\mathcal{M}(\mathcal{E}h) = \xi(s + 1/2)/4$ for the literal displayed input; the normalized input is $4\mathcal{E}h$. The residue kernel retains the full summation index and its 2- and 3-Euler-factor restoration. The Leech theta transform includes its Ramanujan term and variable $w = 12s + 6$. Kernel approximation does not prove the comparison with a Weil eigenfunction. No new analytic existence or zero-location result is asserted by these replays. The source is Connes–Consani–Moscovici, Section 7 and the separately stated subsequent comparison.

The Terzo exponential evaluation has its stated Schanuel-dependent injectivity claim. The unconditional period-subgroup embedding is a different assertion. No symbolic return in the tensor/lattice code turns the conditional whole-ring evaluation into an unconditional injection.

5. TWO GENUINE OUTPUT-VALIDATION REPAIRS

The inherited mathematics certificates pass unchanged, but the inverse command-line utility had two narrower validation gaps. In four-view mode it checked membership and only the first trace triple. It therefore accepted a deliberately doubled fourth right-inverse column, whose first trace is zero but whose later readings were wrong. In shortest mode it trusted a representative's claimed norm metadata after checking membership and trace. Adding twice a nonzero kernel column produced a nonminimal accepted representative.

The replacement `reconstruct_checked.py` checks every external column against the full nine-coordinate right-inverse identity, recomputes all twelve time views, checks the frame convention, verifies the selected residue representative's actual reduced trace and norm, and verifies the output against the proved shortest-norm formula. Two mutation tests establish that the old program accepts the deliberately bad data and the replacement rejects it. Unchanged inputs return the same intended values normally and with Python optimization. These are repairs of validation coverage, not counterexamples to the original mathematical lift theorem.

The standalone short-height proof below additionally reduces attainment from a 196,560-vector census to eight explicit norm-four seeds, one norm-six dodecad, and exact phase/sign transports. This is a proof simplification; it does not multiply the number of independent reviews.

6. SOURCE, NOVELTY, AND CONTINUATION BOUNDARY

The source registry separates imported constructions, independently checked coordinate consequences, new composition witnesses, and still-unproved existence arrows. Content-level searches covered octad integral spans and Smith invariants, three-octad complements and norm counts, octonionic/MOG comparisons, and the existing Fable/Albert constructions. In particular Wilson's existing model isometry is acknowledged explicitly. No absence-from-literature claim is inferred from a failed search.

Three short, self-contained research drafts isolate the full-symmetry integral trace formula, the aligned model isometry with its cubic correction, and the shortest-lift theorem. Each has its own definitions, proof and finite certificate, rather than referring a reader to hundreds of pages for an unstated premise. They are research drafts, not published or independently refereed results.

Nothing here asserts universal positive ES occupancy. What has improved is the exactness of the maps on which a future arithmetic argument could rely: both symmetry generators now have a lossless linear model; the integral image has an explicit syndrome; the two Leech models have a specified inverse isometry; and the discrepancy between trace arithmetic and the full cubic is a retained polynomial, not an omitted term.

Short proof modules

7. THE FIXED COORDINATE MODEL

Write $\Omega = \mathbb{F}_{23} \cup \{\infty\}$, serializing ∞ as coordinate 23. If Q is the set of nonzero squares modulo 23, let $\mathcal{C} \subset \mathbb{F}_2^{24}$ be the span of the 23 supports $(a + Q) \cup \{\infty\}$. A word is represented by the binary integer whose i th bit is its i th coordinate. The twelve words

$$\begin{aligned} &85335e, 5606be, 2ca9ae, 11fe26, 0f55e2, 046f3e, \\ &02379f, 0149a6, 00a4d3, 0042de, 003e4a, 001f25 \end{aligned}$$

form a basis; the notation is hexadecimal. Row reduction of the defining supports gives this basis. Enumerating its 2^{12} sums gives weights 0, 8, 12, 16, 24 with multiplicities 1, 759, 2576, 759, 1. This is the projective Golay code [Chapman(1997)]. We use the Leech model [Brouwer(n.d.), Theorem 1.4]

$$\Lambda = \left\{ x/\sqrt{8} : \begin{array}{l} x \in \mathbb{Z}^{24}, \quad x_i \equiv m \pmod{2} \text{ for one } m \in \{0, 1\}, \\ \sum_i x_i \equiv 4m \pmod{8}, \quad ((x_i - m)/2 \bmod 2)_i \in \mathcal{C} \end{array} \right\}. \quad (1)$$

Thus Λ is even, unimodular and has minimum squared norm four. These properties and the lattice itself are inherited, not claimed as new.

The three disjoint octads are

$$\begin{aligned} B_0 &= \{0, 1, 3, 4, 7, 8, 16, 23\}, \\ B_1 &= \{10, 12, 15, 18, 19, 20, 21, 22\}, \\ B_2 &= \{2, 5, 6, 9, 11, 13, 14, 17\}. \end{aligned}$$

Put $w_i = 21_{B_i}/\sqrt{8}$ and $P_i(x/\sqrt{8}) = \sum_{j \in B_i} x_j/4$. The matrices

$$C = \begin{pmatrix} 12 & 5 \\ 1 & 12 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 7 \\ 3 & 22 \end{pmatrix}, \quad T = \begin{pmatrix} 4 & 11 \\ 16 & 4 \end{pmatrix}$$

act on Ω by fractional linear transformations over $\mathbb{F}_{23}$ and permute lattice coordinates. They preserve $\mathcal{C}$; this can be checked on its twelve basis words. In projective action $T^4 = C$, and C cycles B_0, B_1, B_2 . All vector actions below are these coordinate permutations, not octonion-coordinate permutations.

8. THE SMALLEST STABLE LINEAR COMPLETION

For $n = 0, 1, 2, 3$ retain $P(T^n \lambda)$. Their sums agree, so they give nine independent integer readouts, denoted P_9 . Specifically, retain the first triple and the first two entries of each later triple.

Proposition 1. *The kernel of P_9 is not J -stable. The linear span of its nine functionals and their J -translates has dimension 15. The smallest $\langle J, T \rangle$ -stable real span containing them has dimension 24.*

Proof. The lattice vector with numerator

$$v = (8, -8, 8, 0, -8, 0, -8, 0, 8, 0, \dots, 0)$$

has $P_9(v/\sqrt{8}) = 0$ but

$$P_9(Jv/\sqrt{8}) = (0, 0, 0, -2, -2, 2, 0, 0, 2).$$

The code condition is trivial and the numerator sum is zero. Substitution verifies both readouts; rational row reduction gives ranks 9 and 15. For completeness of the stable span, act with J, T on B_0 , retaining every distinct resulting support. This bounded calculation produces exactly the 759 weight-eight words of $\mathcal{C}$. The next theorem proves that their indicator functionals span rank 24. This proves minimality, not just the existence of one 24-dimensional completion. The enumeration rule is complete: begin with B_0 , apply both generators to every new support, and stop only when the list is closed. Both generators are invertible on the finite set. $\square$

Let $\mathcal{O}$ be the set of all 759 octads. For an arbitrary real vector x , put

$$a_B = \frac{1}{4} \sum_{i \in B} x_i, \quad S = \sum_{B \in \mathcal{O}} a_B, \quad S_i = \sum_{B \ni i} a_B.$$

Theorem 2 (Complete octad reconstruction). *The map $x \mapsto (a_B)_{B \in \mathcal{O}}$ is injective and has inverse on its image*

$$x_i = \frac{23S_i - 7S}{1012}. \quad (2)$$

Its real image consists exactly of data satisfying $a_B = \frac{1}{4} \sum_{i \in B} x_i(a)$ for all B . This is an explicit set of linear equations, not an existence condition.

Proof. Let A be the 759×24 indicator matrix. The Golay octads form the Steiner system $S(5, 8, 24)$ [Chapman(1997), Theorem 1]. Each coordinate lies in 253 octads and each pair in 77, by counting five-subsets. Hence

$$A^T A = 176I + 77\mathbf{1}\mathbf{1}^T.$$

In particular its rank is 24. Directly, $S = 253 \sum x_i/4$ and $S_i = 44x_i + 77 \sum x_j/4$; elimination yields (2). Applying the forward map tests precisely the displayed image equations. Both compositions are the identity on their stated domains. $\square$

9. EXACTLY ONE CYCLIC INTEGRAL OBSTRUCTION

Let $L_{\mathcal{O}}$ be the integral lattice generated by $2\mathbf{1}_B/\sqrt{8}$, for all $B \in \mathcal{O}$.

Lemma 3.

$$L_{\mathcal{O}} = \{x/\sqrt{8} \in \Lambda : \sum_i x_i \equiv 0 \pmod{16}\}, \quad [\Lambda : L_{\mathcal{O}}] = 4.$$

Moreover, with $v = \mathbf{1}/(2\sqrt{8})$,

$$L_{\mathcal{O}}^* = \Lambda + \mathbb{Z}v, \quad L_{\mathcal{O}}^*/\Lambda \cong \mathbb{Z}/4\mathbb{Z}.$$

Proof. Every generator satisfies the displayed congruence. On Λ the sum is divisible by four, and $x \mapsto \sum x_i/4 \pmod{4}$ is onto: the vector $(-3, 1, \dots, 1)/\sqrt{8}$ maps to 1. Thus the congruence sublattice has index four.

Here is a small, fully specified determinant certificate for the reverse inclusion. Sort the octad masks as increasing binary integers, numbered from zero. The 24 rows at positions

$$\begin{aligned} &197, 555, 724, 671, 582, 487, 580, 464, 333, 59, 49, 397, \\ &55, 427, 249, 304, 329, 72, 195, 630, 672, 16, 180, 402 \end{aligned}$$

have determinant 2^{14} . This is checked by exact fraction-free elimination. After scaling every row by $2/\sqrt{8}$, their covolume is $2^{14}/2^{12} = 4$. They generate a sublattice of the congruence sublattice with the same covolume, so equality holds.

Since Λ is integral, $\Lambda \subset L_{\mathcal{O}}^*$. Also $(v, 2\mathbf{1}_B/\sqrt{8}) = 1$, so $v \in L_{\mathcal{O}}^*$. The vector $4v$ lies in Λ , but $2v$ does not, by (1). The quotient $L_{\mathcal{O}}^*/\Lambda$ has order four by unimodularity and the index just proved. Hence v generates it. $\square$

Theorem 4 (Integral gluing syndrome). *Suppose $a \in \mathbb{Z}^{\mathcal{O}}$ satisfies the real image equations of Theorem 2. Then*

$$\delta(a) = \frac{11S - 23S_0}{506}$$

is an integer. There is a unique $\lambda = x/\sqrt{8} \in \Lambda$ having these octad traces if and only if

$$\boxed{\delta(a) \equiv 0 \pmod{4}.} \tag{3}$$

When it exists, (2) reconstructs all its coordinates. For example, the constant trace assignment $a_B = 1$ is linearly compatible but has $\delta = 5$, so it is not a Leech trace packet.

Proof. Compatibility gives a real vector pairing integrally with all generators of $L_{\mathcal{O}}$, hence an element of $L_{\mathcal{O}}^* = \Lambda + \mathbb{Z}v$. Set $v_0 = (-3, 1, \dots, 1)/\sqrt{8} \in \Lambda$. The homomorphism $z \mapsto 4(z, v_0) \pmod{4}$ vanishes on Λ and sends v to 5 mod 4, so it identifies that order-four quotient. In raw coordinates it is

$$4(x/\sqrt{8}, v_0) = (\sum_i x_i - 4x_0)/2 = (11S - 23S_0)/506.$$

This proves integrality, the criterion, and uniqueness. Constant traces reconstruct $x_i = 1/2$, that is v ; their syndrome is five. $\square$

10. SCOPE AND VERIFICATION

The output is a complete integral coordinate theorem and a minimal linear symmetry completion. It does not assert that the additional Erdős–Straus cubic contains a positive integral trace. The data and maps are classical-code constructions; no new lattice, uniqueness theorem, or historical-priority claim is made. A content-level search of Golay design lattices, Smith invariants, octad trios and octonionic models did not establish a prior exact statement of this combined marked reconstruction; it also cannot establish absence from the literature.

The standalone `verify_audit.py` reconstructs the code, the generator orbit, all entries of $A^T A$, the determinant minor, the kernel witness, and exact inverse/syndrome tests. The finite facts entering the proof have fixed bounds (4096 codewords, 759 octads, a 24×24 determinant). No floating point, unbounded

search, formal-prover run or independent reviewer is used. The general assertions follow from the arguments above, not from extrapolating these tests.

11. THE FIXED COORDINATE MODEL

Write $\Omega = \mathbb{F}_{23} \cup \{\infty\}$, serializing ∞ as coordinate 23. If Q is the set of nonzero squares modulo 23, let $\mathcal{C} \subset \mathbb{F}_2^{24}$ be the span of the 23 supports $(a + Q) \cup \{\infty\}$. A word is represented by the binary integer whose i th bit is its i th coordinate. The twelve words

$$\begin{aligned} &85335e, 5606be, 2ca9ae, 11fe26, 0f55e2, 046f3e, \\ &02379f, 0149a6, 00a4d3, 0042de, 003e4a, 001f25 \end{aligned}$$

form a basis; the notation is hexadecimal. Row reduction of the defining supports gives this basis. Enumerating its 2^{12} sums gives weights 0, 8, 12, 16, 24 with multiplicities 1, 759, 2576, 759, 1. This is the projective Golay code [Chapman(1997)]. We use the Leech model [Brouwer(n.d.), Theorem 1.4]

$$\Lambda = \left\{ x/\sqrt{8} : \begin{array}{l} x \in \mathbb{Z}^{24}, \quad x_i \equiv m \pmod{2} \text{ for one } m \in \{0, 1\}, \\ \sum_i x_i \equiv 4m \pmod{8}, \quad ((x_i - m)/2 \bmod 2)_i \in \mathcal{C} \end{array} \right\}. \quad (4)$$

Thus Λ is even, unimodular and has minimum squared norm four. These properties and the lattice itself are inherited, not claimed as new.

The three disjoint octads are

$$\begin{aligned} B_0 &= \{0, 1, 3, 4, 7, 8, 16, 23\}, \\ B_1 &= \{10, 12, 15, 18, 19, 20, 21, 22\}, \\ B_2 &= \{2, 5, 6, 9, 11, 13, 14, 17\}. \end{aligned}$$

Put $w_i = 2\mathbf{1}_{B_i}/\sqrt{8}$ and $P_i(x/\sqrt{8}) = \sum_{j \in B_i} x_j/4$. The matrices

$$C = \begin{pmatrix} 12 & 5 \\ 1 & 12 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 7 \\ 3 & 22 \end{pmatrix}, \quad T = \begin{pmatrix} 4 & 11 \\ 16 & 4 \end{pmatrix}$$

act on Ω by fractional linear transformations over $\mathbb{F}_{23}$ and permute lattice coordinates. They preserve $\mathcal{C}$; this can be checked on its twelve basis words. In projective action $T^4 = C$, and C cycles B_0, B_1, B_2 . All vector actions below are these coordinate permutations, not octonion-coordinate permutations.

12. AN OCTAD-ALIGNED COORDINATE ISOMETRY

Use Wilson's octonion basis $(1, i_0, \dots, i_6)$, with $i_0 i_1 = i_3$, its translates modulo seven, and anticommutativity. Thus $i_t i_{t+1} = i_{t+3}$ and each $i_t^2 = -1$. Let $L \subset \mathbb{O}$ be the E_8 lattice generated by $\pm i_t \pm i_u$ and the half-sums having an odd number of minus signs; here $i_\infty = 1$. Put

$$s = (-1 + i_0 + \dots + i_6)/2.$$

Wilson's Leech lattice consists of triples $z = (z_0, z_1, z_2)$ with

$$z_i \in L, \quad z_i + z_j \in L\bar{s}, \quad z_0 + z_1 + z_2 \in Ls, \quad \|z\|_W^2 = \frac{1}{2} \sum_i N(z_i). \quad (5)$$

The bar, side of multiplication and norm factor are part of the definition [Wilson(2009), Sections 2–4].

Theorem 5 (Specified model isometry). *The following signed coordinate permutation identifies (4) with (5), and aligns the three phase octads with the three octonions:*

j	$\pi(8j), \dots, \pi(8j+7)$								$\epsilon(8j), \dots, \epsilon(8j+7)$							
0	0	1	3	4	8	23	7	16	+	−	−	−	−	−	−	−
1	10	22	15	18	12	21	19	20	−	−	−	−	+	+	−	+
2	2	13	5	11	6	14	17	9	−	−	−	−	+	+	−	+

For $\lambda = x/\sqrt{8}$, its image has raw coordinates

$$y_i = \epsilon_i x_{\pi(i)}, \quad z_j = \frac{1}{2} \sum_{r=0}^7 y_{8j+r} e_r, \quad (e_0, e_1, \dots, e_7) = (1, i_0, \dots, i_6).$$

The inverse is $x_{\pi(i)} = \epsilon_i y_i$. Both maps are integral in their stated numerator coordinates and preserve squared norm.

Proof. The signed permutation is invertible and $\sum y_i^2/8 = \sum x_i^2/8$, proving the metric statement. For lattice membership it suffices to test these 37 explicit generators of (4):

$$4(e_i - e_{23}) \ (0 \leq i < 23), \quad 8e_{23}, \quad 2c \ (c \text{ one of the twelve binary basis words}), \quad (-3, 1, \dots, 1).$$

They are numerator vectors. To prove generation, the even part is generated by $4D_{24}$ and the twelve $2c$: binary addition differs from integer addition by four times an intersection vector, and those intersections

have even weight because the code is self-orthogonal. The first 24 displayed vectors generate $4D_{24}$. Any two odd lattice vectors differ by an even one, so the final vector supplies the odd coset.

Here are exact integer tests for each image, permitting the 37 verifications without any geometric judgement. A block $v = 2z$ represents a member of L precisely when all entries have common parity m and $\sum v_i \equiv 2m \pmod{4}$. To test $z \in L\bar{s}$, multiply its raw block on the right by $(-1, 1, \dots, 1)$; all product coordinates must be divisible by four and their quotients must pass the block test. To test $z \in Ls$, use $(-1, -1, \dots, -1)$. These rules use $N(s) = 2$ and retain all parentheses. Apply them to the three blocks, their three pair-sums and their total sum. The table above passes all 37 generator tests; the accompanying certificate lists every generator image and the independent checker performs the integer operations from the basis multiplication table.

This proves inclusion in Wilson's lattice. Wilson proves that (5) is unimodular at the displayed scale [Wilson(2009), Sections 3–4]. The source lattice is unimodular, so the isometric inclusion has index one. This upgrades generator checking to equality of the full infinite lattices; a minimal-shell census alone would not suffice. $\square$

Wilson already supplies an octonionic-to-MOG comparison in Section 4. The assertion here is the displayed *projective-octad-aligned* coordinate choice and its arithmetic readout, not the first existence of an isometry between Leech models.

13. THE EXACT ALBERT EMBEDDING AND THE RETAINED CUBIC

The three scalar traces become

$$P_j(\lambda) = \frac{1}{2} \sum_{r=0}^7 \epsilon_{8j+r} (z_j)_r = \frac{1}{2} \operatorname{Re}(z_j \bar{h}_j), \quad h_j = \sum_{r=0}^7 \epsilon_{8j+r} e_r. \quad (6)$$

For the full, injective off-diagonal map put

$$\iota(z) = \begin{pmatrix} 0 & z_2 & \bar{z}_1 \\ \bar{z}_2 & 0 & z_0 \\ z_1 & \bar{z}_0 & 0 \end{pmatrix}.$$

Reading the marked entries is the inverse. The quadratic and cubic invariants are

$$\operatorname{Tr}(\iota(z)^2) = 4\|\lambda\|^2, \quad \mathcal{N}(\lambda) = N_J(\iota(z)) = 2 \operatorname{Re}((z_0 z_1) z_2).$$

The association displayed in the last expression is fixed. In integer numerator coordinates,

$$\mathcal{N}(\lambda) = \frac{1}{4} \operatorname{Re}((y_0 y_1) y_2),$$

where y_i now denotes the corresponding raw octonion block. Expansion has exactly 64 nonzero monomials; their complete signs and coordinate indices are serialized in the certificate.

Proposition 6 (Two failed descents, with witnesses). *The cubic $\mathcal{N}$ does not descend through P , and the specific phase isometry C does not preserve $\mathcal{N}$.*

Proof. The lattice numerator

$$(-4, 2, 2, 0, 0, -2, -2, 2, 2, 0, 2, 0, 0, 2, 0, 2, 2, 0, 0, -2, -2, 0, 0, -4)$$

has $P = 0$ and $\mathcal{N} = -8$, whereas $\mathcal{N}(0) = 0$. For

$$b_0 = \frac{1}{\sqrt{8}}(-3, 1, -1, 1, 1, -1, -1, 1, 1, 1, 1, 1, -1, 1, 1, 1, -1, -1, -1, -1, 1, 1)$$

one computes

$$\mathcal{N}(b_0) = -5, \quad \mathcal{N}(Cb_0) = 1, \quad \mathcal{N}(C^2 b_0) = -1.$$

Every assertion is direct lattice membership, a block-sum calculation, or multiplication in the stated octonion table. $\square$

This does not prohibit a transported cubic: g preserves the marked pair consisting of the lattice and the function $\mathcal{N} \circ g^{-1}$. It prohibits dropping that function's change of coordinates. The cyclic permutation of the three Wilson blocks is a different operator from the projective phase permutation C .

14. A FAITHFUL COMBINED STATE AND ITS EXACT ES CORRECTION

Let $t = P\lambda$, $n_i = N(z_i)$ and $\nu = 2 \operatorname{Re}((z_0 z_1) z_2)$. Retain the full state

$$\hat{X}(\lambda) = \begin{pmatrix} t_0 & z_2 & \bar{z}_1 \\ \bar{z}_2 & t_1 & z_0 \\ z_1 & \bar{z}_0 & t_2 \end{pmatrix}.$$

Its image is specified by the lattice condition on z and the three linear diagonal constraints (6); its inverse reads z and applies Theorem 5. The Albert formulas give

$$N_J(\hat{X}) = t_0 t_1 t_2 - \sum_i t_i n_i + \nu, \quad S_J(\hat{X}) = \sum_{i < j} t_i t_j - \sum_i n_i.$$

Thus, for $F_p(t) = 4t_0 t_1 t_2 - p \sum_{i < j} t_i t_j$,

$$\boxed{4N_J(\hat{X}) - pS_J(\hat{X}) = F_p(t) + \sum_i (p - 4t_i)n_i + 4\nu.} \quad (7)$$

Every mixed term is necessary. A norm-preserving transformation of the ambient Albert algebra is not automatically a map of this lattice graph or of the positive ES locus. Conversely, a lattice isometry need not preserve the fixed cubic, as Proposition 6 shows. Formula (7) is the corrected interface, not a claim that these structures are unrelated.

15. VERIFICATION AND SCOPE

The independent standard-library checker verifies all 37 generator images and their inverses, and the compact-Zorn/three-block return over exact $\mathbb{Q}(i)$ coefficients. It tests the cubic identity on 2600 determining nodes: the 24 basis vectors, $e_i + e_j$ and $e_i - e_j$ for all pairs, and $e_i + e_j + e_k$ for all distinct triples. For a homogeneous cubic these separate every coefficient; testing only $e_i + e_j$ would not separate $x_i^2 x_j$ from $x_i x_j^2$.

The note provides an explicit convention-compatible isometry and corrected polynomial interface. It asserts neither a new Leech lattice nor a solution of Erdős–Straus. Content-level literature searches are recorded with the source package; historical priority for these particular coordinate formulas is not claimed. No theorem prover or independent reviewer was used.

16. THE FIXED COORDINATE MODEL

Write $\Omega = \mathbb{F}_{23} \cup \{\infty\}$, serializing ∞ as coordinate 23. If Q is the set of nonzero squares modulo 23, let $\mathcal{C} \subset \mathbb{F}_2^{24}$ be the span of the 23 supports $(a + Q) \cup \{\infty\}$. A word is represented by the binary integer whose i th bit is its i th coordinate. The twelve words

$$\begin{aligned} &85335e, 5606be, 2ca9ae, 11fe26, 0f55e2, 046f3e, \\ &02379f, 0149a6, 00a4d3, 0042de, 003e4a, 001f25 \end{aligned}$$

form a basis; the notation is hexadecimal. Row reduction of the defining supports gives this basis. Enumerating its 2^{12} sums gives weights 0, 8, 12, 16, 24 with multiplicities 1, 759, 2576, 759, 1. This is the projective Golay code [Chapman(1997)]. We use the Leech model [Brouwer(n.d.), Theorem 1.4]

$$\Lambda = \left\{ x/\sqrt{8} : \begin{array}{l} x \in \mathbb{Z}^{24}, \quad x_i \equiv m \pmod{2} \text{ for one } m \in \{0, 1\}, \\ \sum_i x_i \equiv 4m \pmod{8}, \quad ((x_i - m)/2 \bmod 2)_i \in \mathcal{C} \end{array} \right\}. \quad (8)$$

Thus Λ is even, unimodular and has minimum squared norm four. These properties and the lattice itself are inherited, not claimed as new.

The three disjoint octads are

$$\begin{aligned} B_0 &= \{0, 1, 3, 4, 7, 8, 16, 23\}, \\ B_1 &= \{10, 12, 15, 18, 19, 20, 21, 22\}, \\ B_2 &= \{2, 5, 6, 9, 11, 13, 14, 17\}. \end{aligned}$$

Put $w_i = 2\mathbf{1}_{B_i}/\sqrt{8}$ and $P_i(x/\sqrt{8}) = \sum_{j \in B_i} x_j/4$. The matrices

$$C = \begin{pmatrix} 12 & 5 \\ 1 & 12 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 7 \\ 3 & 22 \end{pmatrix}, \quad T = \begin{pmatrix} 4 & 11 \\ 16 & 4 \end{pmatrix}$$

act on Ω by fractional linear transformations over $\mathbb{F}_{23}$ and permute lattice coordinates. They preserve $\mathcal{C}$; this can be checked on its twelve basis words. In projective action $T^4 = C$, and C cycles B_0, B_1, B_2 . All vector actions below are these coordinate permutations, not octonion-coordinate permutations.

17. THE SHORTEST INVERSE

For $r \in \{0, 1, 2, 3\}^3$, set $a_i(r) = r_i$ for $r_i \leq 2$ and $a_i(r) = -1$ for $r_i = 3$. Put $d(r) = \sum_i a_i(r)^2$.

Theorem 7. *Every $t \in \mathbb{Z}^3$ is $P\lambda$ for some $\lambda \in \Lambda$, and*

$$\boxed{\min_{P\lambda=t} \|\lambda\|^2 = \frac{\|t\|^2}{4} + \mu(t \bmod 4)}, \quad \mu(r) = \begin{cases} 0 & r = 000, \\ 3 & r = 222, \\ 4 - d(r)/4 & \text{otherwise.} \end{cases}$$

In particular $0 \leq \mu(r) \leq 15/4$. The inverse below attains this minimum at every integer input, including negative and zero traces.

Proof. We give eight norm-four seeds, rather than require a reader to trust a census of all minimal vectors. For subsets $N \subset S \subset \Omega$ define the numerator $E(S, N) = 2\mathbf{1}_S - 4\mathbf{1}_N$. Define $O(j, \epsilon, N)$ to have coordinate 3ϵ at j and $1 - 2\mathbf{1}_N(i)$ elsewhere. Entries in the following table are numerators, to be divided by $\sqrt{8}$:

Trace	Seed
100	$O(0, -1, \{2, 5, 6, 10, 12, 13, 15, 22\})$
110	$E(\{0, 2, 7, 11, 12, 16, 19, 23\}, \{0, 2\})$
111	$O(23, 1, \{0, 1, 2, 3, 9, 12, 21\})$
200	$E(\{0, 1, 3, 4, 7, 8, 16, 23\}, \{0, 1\})$
210	$O(16, 1, \{0, 2, 5, 6, 11, 18, 22\})$
211	$E(\{7, 8, 12, 13, 14, 16, 18, 23\}, \emptyset)$
220	$E(\{4, 7, 16, 18, 19, 20, 21, 23\}, \emptyset)$
221	$O(2, -1, \emptyset)$

Each seed satisfies (8), has exactly the indicated trace and squared norm four; these are eight explicit integer/code calculations. The matrix

$$R = \begin{pmatrix} 5 & 6 \\ 19 & 0 \end{pmatrix} \in \text{SL}_2(\mathbb{F}_{23})$$

preserves the code and exchanges B_1, B_2 while fixing B_0 . Together with C it realizes all permutations of the three traces. Membership of $RC = \mathcal{C}$ is verified on the twelve code basis words.

Independently changing the signs on each entire block also preserves (8). Indeed, write $x = m\mathbf{1} + 2c + 4z$. Self-orthogonality gives $|c \cap B_j|$ even; hence $\sum_{i \in B_j} x_i$ is divisible by four, and changing those signs preserves the numerator-sum congruence. The binary word changes from c to $c + m\mathbf{1}_{B_j}$, still a codeword. The squared norm is unchanged. Thus permutations and block signs transport the eight seeds to all 62 nonzero reduced traces except 222.

Every nonzero vector of Λ has squared norm at least four, proving optimality at those 62 reduced traces. At 000 use zero. At 222, the numerator sum is 24, so the common parity in (8) is even. Write $x = 2y$. Each block sum of y is four, and

$$\|x/\sqrt{8}\|^2 = \frac{1}{2} \sum_i y_i^2 \geq \frac{1}{2} \sum_i |y_i| \geq 6.$$

Equality is attained by $E(S, \emptyset)/\sqrt{8}$ for

$$S = \{0, 3, 4, 5, 6, 8, 9, 10, 12, 13, 15, 18\},$$

a Golay dodecad with four entries in each block. This proves the reduced minimum six.

Finally, $(w_i, w_j) = 4\delta_{ij}$. The component of any vector in the real span of the w_i is

$$q(t) = \frac{1}{4} \sum_i t_i w_i, \quad \|q(t)\|^2 = \|t\|^2/4.$$

Translation by w_i changes only the i th trace by four and leaves the perpendicular component unchanged. For a reduced representative ν_r just constructed, the vector

$$\lambda_*(t) = \nu_r + \sum_i \frac{t_i - a_i(r)}{4} w_i, \quad r = t \bmod 4$$

is integral, has trace t , and has the same minimal perpendicular norm as ν_r . This proves the all-input formula and both the asserted minimum and its attainment. $\square$

18. FIBRES, RETAINED STATE AND ARITHMETIC SCOPE

The three norm-four vectors b_0, Cb_0, C^2b_0 , with

$$b_0 = \frac{1}{\sqrt{8}}(-3, 1, -1, 1, 1, -1, -1, 1, 1, 1, 1, 1, -1, 1, 1, 1, -1, -1, -1, -1, 1, 1),$$

have respective traces $(1, 0, 0), (0, 1, 0), (0, 0, 1)$. Put $\sigma(t) = \sum_i t_i C^i b_0$ and $K = \ker P$. Then

$$(t, k) \mapsto \sigma(t) + k, \quad \lambda \mapsto (P\lambda, \lambda - \sigma(P\lambda))$$

are exact inverse group homomorphisms between $\mathbb{Z}^3 \oplus K$ and Λ . This is not an orthogonal direct product. The rank of K is 21 and its determinant is 64: $L = \bigoplus_i \mathbb{Z} w_i$ is primitive by pairing with the $C^i b_0$, and $\Lambda/(L \oplus K) \cong (\mathbb{Z}/4)^3$, so unimodularity gives $\det L \det K = 64^2$ with $\det L = 64$.

For a symmetry g descending to ρ_g on the traces, the full inverse action is

$$(t, k) \mapsto (\rho_g t, gk + g\sigma(t) - \sigma(\rho_g t)).$$

The kernel correction cannot be omitted merely because a shortest representative was selected. If a symmetry does not descend to the traces, the full coordinate vector or a larger stable readout is required.

For $F_p(t) = 4t_0 t_1 t_2 - p \sum_{i < j} t_i t_j$, surjectivity identifies positive integral ES solutions with lattice vectors satisfying $P\lambda > 0$ and $F_p(P\lambda) = 0$. It does not prove that this additional cubic locus is occupied. One immediate exact restriction is $r \neq 222$ for odd p : that residue gives $F_p(t) \equiv 4 \pmod{8}$.

19. REPRODUCIBILITY AND ATTRIBUTION

`verify_shortest.py`, using only Python's standard library and the accompanying coordinate routines, verifies the eight seeds, the dodecad, both phase generators, and all 64 transported representatives. The all-integer conclusion uses translation and the Leech minimum, not enumeration up to an unspecified height. `reconstruct_checked.py` also verifies the returned vector's exact height against the formula before describing it as shortest.

The Leech model, Golay code and their symmetries are inherited [Brouwer(n.d.), Chapman(1997)]; Wilson's three-octonion construction is related background [Wilson(2009)]. This note isolates the explicit three-trace height formula and provides a shorter seed proof than the preceding full-shell computation. A scoped content search did not identify an exact earlier statement of this formula, but no historical-priority or new-lattice claim is made. No formal theorem-prover build or independent review is claimed.

20. THE COMPLETE SOURCE-ADDRESSABLE ARROW REGISTER

Full file hashes and exact formula strings are in `morphisms.json`; the prose version is `MORPHISMS.md`. Each entry states the source, target, domain, map, return, fibre and information boundary.

M01 – Divisor to primitive marked ray. Source. $a > 0, u | a^2$

Target. (h, r, s) , $\gcd(r, s) = 1$

Domain. positive integers; all available prime exponents $0..2v_q(a)$

Map. $d = \gcd(a, u)$; $r = u/d$; $s = a/d$; $h = d/r$

Return. $a = hrs$, $u = hr^2$

Fibre. singleton

Retained information. None. Do not replace the exponent box by the subgroup it generates.

Status. retained

Evidence. written source and bounded original replay

Locators. `inputs/erdos_straus_research/paper.tex:89`, `prop:divisor`.

M02 – Exterior denominator reconstruction. Source. $(p, R, u, h, r, s, E, \kappa)$

Target. ordered (x, y, z) with retained p, R, E

Domain. p odd prime; $0 < R < p$; $R | pr + s$; $p + R = 4hrs$; $\kappa = (pr + s)/R$

Map. $(hrs, hs * \kappa, p * h * r * \kappa)$

Return. $u = x^2 / (R * y - p * x)$, then M01; κ from p, r, s, R

Fibre. singleton on the marked image

Retained information. Sorting/permuting is not part of this arrow.

Status. retained

Evidence. written source and bounded original replay

Locators. `inputs/erdos_straus_research/paper.tex:75`, `eq:E`; `inputs/erdos_straus_research/paper.tex:100`, `eq:Ediv`.

M03 – Middle denominator reconstruction. Source. $(p, R, u, h, r, s, M, \lambda)$

Target. ordered (x, y, z) with retained p, R, M

Domain. p odd prime; $0 < R < p$; $R | r + s$; $p + R = 4hrs$; $\lambda = (r + s)/R$

Map. $(hrs, p * h * s * \lambda, p * h * r * \lambda)$

Return. $u = p * x^2 / (R * y - p * x)$, then M01

Fibre. singleton on the marked image

Retained information. λ is not exterior κ .

Status. retained

Evidence. written source and bounded original replay

Locators. `inputs/erdos_straus_research/paper.tex:83`, `eq:M`; `inputs/erdos_straus_research/paper.tex:101`, `eq:Mdiv`.

M04 – Middle orientation quotient. Source. oriented middle divisor states

Target. increasing denominator triples

Domain. same bounded domain as M03; R odd > 1

Map. $u \rightarrow$ min-order of the last two denominators

Return. retain u or an orientation bit to invert

Fibre. exactly two, u and a^2/u

Retained information. The two marked states must not be counted as two distinct sorted triples.

Status. retained

Evidence. written source and bounded original replay

Locators. `inputs/erdos_straus_research/paper.tex:128`, `prop:orientation`; `inputs/erdos_straus_research/paper.tex:133`, `eq:middleinvolution`.

M05 – Prime-incidence root blocks. Source. eligible finite auxiliary-prime set and rays

Target. finite joint incidence law

Domain. auxiliary q does not divide coefficients; determinant-collision primes retained; all cutoffs fixed for PNT

Map. ray $(r, s) \rightarrow -s/r \bmod q$; group equal roots

Return. Retain ray labels and the entire root block

Fibre. a block may contain several rays

Retained information. A density result is not an inverse returning a state at each prime.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/erdos_straus_research/paper.tex:193, thm:pgf; inputs/erdos_straus_research/paper.tex:533, prop:collisions.

M06 – Composite residual extraction. Source. actual prime-factor occurrences in a residual

Target. bounded Omega residual

Domain. original bounded exponent multiset; quotient-group zero-sum bounds 2,3,5

Map. retain a minimal target-product submultiset

Return. No unique inverse: record deleted occurrences and the source residual

Fibre. all extensions by discarded occurrences satisfying the divisor budget

Retained information. Existence of a residue in generated support is insufficient.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/erdos_straus_research/paper.tex:259, lem:extract; inputs/erdos_straus_research/paper.tex:266, thm:235.

M07 – Same-shell unimodular companion map. Source. state $v=(r,s)$, companion $w=(b,d)$

Target. all same- (p,R) exterior rays

Domain. $rd-sb=\epsilon+1$; A, RB primitive; new coordinates positive; $r_prime*s_prime|a$

Map. $v_prime=A*v+R*B*w$; $kappa_prime=A*kappa+B*(bp+d)$

Return. $A=\epsilon*(d*r_prime-b*s_prime)$; $B=\epsilon*(r*s_prime-s*r_prime)/R$

Fibre. singleton for chosen basis

Retained information. Retain $p,R,basis$ and shell divisor test. The ordinary mediant does not preserve $R>1$.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/erdos_straus_research/paper.tex:548, thm:lattice; inputs/erdos_straus_research/paper.tex:555, eq:latticemap.

M08 – Exterior-to-middle cubic exchange. Source. exterior marking

Target. middle marking plus square-divisor index g

Domain. $g=\gcd(h,r)$; necessary and sufficient $R|4r^3-1$

Map. $(H,a1,b1)=(s*g^2,r/g,h/g)$

Return. for each g : $h=g*b1, r=g*a1, s=H/g^2$; require $g^2|H$, coprimality, cubic congruence

Fibre. exact finite g -indexed inverse fibre

Retained information. Forgetting g can identify two distinct exterior states.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/erdos_straus_research/paper.tex:581, thm:cubic; inputs/erdos_straus_research/paper.tex:600, eq:cubicfibre.

M09 – Angular anchor packet. Source. $(p,R,A;t,n,d)$ from actual divisor boxes

Target. positive middle state

Domain. $t|a/A, nd|A, \gcd(n,d)=1$; $tn+d=0 \pmod R$; $tn>8d$; $\gcd(a,R)=1$

Map. $g=\gcd(d,tn)$; $(h,r,s)=(a*g^2/(dtn),d/g,tn/g)$; $u=a*d/(tn)$

Return. $t=d*s/(n*r)$, with original anchor pair retained

Fibre. all anchor pairs giving integral t ; collisions $t*n*d_prime=t_prime*n_prime*d$

Retained information. No unbounded exponent lift or positivity surrogate.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/es_pointwise_tranche/tranche.tex:127, eq:packetmap; inputs/es_pointwise_tranche/tranche.tex:163, eq:packetinverse.

M10 – Exterior shell/hyperbola exchange. Source. (p,R,a,u,E)

Target. (p,k,D,u)

Domain. $k=(4u+1)/R$; all entries positive; $R<p$; $u|a^2$

Map. $k=(4u+1)/R$; $D=(kp+1)/4=k*a-u$

Return. $R=(4u+1)/k$; $a=(u+D)/k$

Fibre. singleton on the stated congruence/divisor locus

Retained information. k differs from $kappa$. Search termination does not imply a nonempty output.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/es_pointwise_tranche/tranche.tex:464, prop:hyperbola.

M11 – Ordered Fourier coordinates. Source. ordered real triple q

Target. (t, ζ) in R direct-sum C

Domain. $\omega = e^{(2\pi i/3)}$; retain complex conjugation and slot order

Map. $t = \sum(q)/3$; $\zeta = (q_0 + \omega q_1 + \omega^2 q_2)/3$

Return. $q_j = t + \omega^{(-j)} \zeta + \omega^j \overline{\zeta}$

Fibre. singleton

Retained information. Do not identify vector-space coordinates with componentwise direct-product multiplication.

Status. retained

Evidence. written source and bounded original replay

Locators. `inputs/three_coordinate_jordan/note.tex:30, eq:fourier; inputs/three_coordinate_jordan/note.tex:35, eq:inverse-fourier.`

M12 – Transported cubic algebra. Source. ordered componentwise multiplication

Target. star multiplication in Fourier coordinates

Domain. same fields and normalization as M11

Map. $(t, z) * (u, w) = (tu + 2\operatorname{Re}(z \overline{w}), t * w + u * z + \overline{z} * \overline{w})$

Return. M11 transports multiplication both ways

Fibre. isomorphism

Retained information. Norm is cubic product; spin-factor $c=0$ is a different product.

Status. retained

Evidence. written source and bounded original replay

Locators. `inputs/three_coordinate_jordan/note.tex:40, prop:product; inputs/three_coordinate_jordan/note.tex:84, eq:family.`

M13 – Rational spin inverse. Source. Q direct-sum $Q(\omega)$, spin product

Target. nonzero elements with Jordan inverse

Domain. $q = t^2 - 2\operatorname{Norm}(z)$ is anisotropic over Q ; not over R

Map. $(t, z) \rightarrow (t, -z)/q$

Return. involution on Jordan-invertible elements

Fibre. singleton

Retained information. Jordan invertibility is not division-algebra cancellation.

Status. retained

Evidence. written source and bounded original replay

Locators. `inputs/three_coordinate_jordan/note.tex:127, thm:rational.`

M14 – Full three-input associator. Source. source algebra $(a, u)(b, v) = (ab + B(u, v), av + bu)$

Target. vector-valued trilinear tensor

Domain. B fixed; output vector space retained

Map. $[a, u; b, v; c, w] = (0, B(u, v)w - B(v, w)u)$

Return. retain every output component, or full family ℓ composed with associator

Fibre. scalar contraction alone has a kernel

Retained information. 27 input cells for a 3D algebra carry 3 output coordinates: 81 scalar coefficients, not one scalar 27-cube.

Status. retained

Evidence. written source and bounded original replay

Locators. `inputs/cube_unification/note.tex:81, prop:recovery.`

M15 – Three marked-face gluing. Source. three bilinear faces with compatible shared edges

Target. scalar $3 \times 3 \times 3$ tensor

Domain. chosen marks e_i and splittings; incompatible shared edges have no lift

Map. inclusion-exclusion lift T_0 plus Q_0 tensor Q_1 tensor Q_2

Return. read faces and subtract T_0

Fibre. affine space of dimension eight

Retained information. Three marked faces do not determine the full cube.

Status. retained

Evidence. written source and bounded original replay

Locators. `inputs/cube_unification/note.tex:173, thm:glue.`

- M16 – Cube kernel-incidence transport. Source.** nondegenerate cube plus three typed planes
Target. three genus-one curves and maps ϕ_{ij}
Domain. rank exactly two along smooth determinant curves; left/right kernels per slot
Map. ϕ_{ij} given by $\text{tensor}(v_i, v_j, -) = 0$
Return. ϕ_{ji} ; retain tensor and all basis changes
Fibre. forgetting tensor or line bundle is not injective
Retained information. Identical cubic equations do not identify incidence maps or full-turn translation.
Status. retained
Evidence. written source and bounded original replay
Locators. `inputs/cube_unification/note.tex:234, lem:kernel; inputs/cube_unification/note.tex:271, thm:infinite.`
- M17 – Arithmetic-face attachment to cube. Source.** existing ordered ES witness plus Moore cube
Target. integral cube, three basis changes, clearing scale
Domain. rank-two arithmetic face; distinct lattice/arithmetic scale conventions
Map. Gaussian normalize faces; $L A R = M$; change all tensor slots and clear denominators
Return. undo each basis change and divide the retained clearing scale
Fibre. singleton with all marks; many cubes without marks
Retained information. This attaches a known witness, not produces an unknown one.
Status. retained
Evidence. written source and bounded original replay
Locators. `inputs/cube_unification/note.tex:716, thm:attachment.`
- M18 – Tensor to graded Lie operator. Source.** T in V_0 tensor V_1 tensor V_2
Target. linear D_T on three Hom blocks
Domain. oriented 3D spaces; first-Tits coefficient field fixed
Map. for pure a^*b^*c : $D(A, B, C) = (b(a \text{ cross } Cc))^T, c(b \text{ cross } Aa)^T, a(c \text{ cross } Bb)^T$
Return. read tensor coefficients from operator entries using ϵ_{ijk}
Fibre. injective
Retained information. $D_u^2 = 0$ and $I + tD_u$ apply to pure u ; not arbitrary sums.
Status. retained
Evidence. written source and bounded original replay
Locators. `inputs/cube_unification/note.tex:361, eq:Du; inputs/cube_unification/note.tex:371, eq:Dinverse; inputs/cube_unification/note.tex:380, thm:move.`
- M19 – First Tits to Zorn-Albert coordinates. Source.** (A, B, C) in three typed Hom blocks
Target. $H_3(\text{split } O)$
Domain. complexification and orientations; fixed multiplication association
Map. $x_1 = (A_{23}, \text{col1 } B, \text{-row1 } C, A_{32})$, cyclic for x_2, x_3 ; $x_i = A_{ii}$
Return. read all matrix entries back
Fibre. linear isomorphism of dimension 27
Retained information. Compact real points satisfy $A = A^*, C = B^*$. Split transformations need not preserve that real form or the chosen lattice.
Status. retained
Evidence. original formal-polynomial replay plus new exact $Q(i)$ determining-grid check
Locators. `inputs/cube_unification/note.tex:495, eq:phi.`
- M20 – Cayley encoding. Source.** ordered ES state $(p; x, y, z)$
Target. $(-p, 4x, 4y, 4z)$ with affine scale
Domain. p positive; denominators positive; $e_3 = 0$ iff ES
Map. $(-p, 4x, 4y, 4z)$
Return. $p = -a_0$, $\text{denominators} = a_i/4$
Fibre. singleton in affine marked coordinates
Retained information. Projectivization forgets a common scale; retain it.
Status. retained
Evidence. written source and bounded original replay
Locators. `inputs/cube_unification/note.tex:784, eq:cayleyes.`
- M21 – Reciprocal Cremona map. Source.** Cayley coordinate torus $e_3 = 0$
Target. projective plane sum $\ell_i = 0$
Domain. every coordinate nonzero
Map. $\ell_i = \text{product}_{\{j \neq i\}} a_j$, equivalently $[1/a_i]$

Return. same product formula; second application scales by $(\text{product } a)^2$

Fibre. singleton projectively, nontrivial scale affinely

Retained information. Coordinate hyperplanes are excluded; exceptional divisors not hidden.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/cayley_fable_bridge/note.tex:223, eq:cremona.

M22 – Four normalized Cayley charts. Source. marked Cayley a and face i

Target. three roots t_{ij} plus normalization h_i

Domain. all a_i nonzero, pairwise distinct; $h_i = -1/(4a_i)$

Map. $t_{ij} = -4a_i/a_j$; normalized $a_i = -1/4$

Return. $a_i = (-1/4)/h_i$, $a_j = 1/(h_i t_{ij})$

Fibre. singleton with face, companion order and scale

Retained information. Changing face can change positivity. Twelve A4 flags, twenty-four S4 flags.

Status. retained

Evidence. 72 all-oriented arithmetic flags independently replayed, 432 local factor returns

Locators. inputs/tetrahedral_markings/note.tex:193, eq:roots; inputs/tetrahedral_markings/note.tex:204, eq:coeffs.

M23 – Oriented face exchange. Source. oriented roots (r,s,t) , sum4

Target. new oriented roots

Domain. $r*s*t$ nonzero; none=-4; all distinct inherited from full torus

Map. $J(r,s,t) = (16/r, -4t/r, -4s/r)$

Return. $J^2 = \text{identity}$

Fibre. singleton

Retained information. Companion swap is necessary for preserved tetrahedral orientation.

Status. retained

Evidence. general rational identities and all 72 arithmetic flags

Locators. inputs/tetrahedral_markings/note.tex:225, eq:J; inputs/tetrahedral_markings/note.tex:231, eq:relations.

M24 – Derivative/companion coordinates. Source. ordered r,s,t

Target. (r,d,η) , $d = f_{\text{prime}}(r)$, $\eta = s - t$

Domain. simple root $d \neq 0$; $\eta^2 = (3r-4)^2 + 16d$

Map. r fixed; $d = -3r^2/4 + 2r + \eta^2/2$; $\eta = s - t$

Return. $s = (4r + \eta)/2$; $t = (4r - \eta)/2$

Fibre. singleton; forgetting η leaves two companion orders

Retained information. η maps to $4\eta/r$ under J ; its square is insufficient for orientation.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/tetrahedral_markings/note.tex:254, eq:rd.

M25 – Ordered roots to cubic coefficients. Source. ordered triple of distinct roots sum4

Target. (α, β, γ)

Domain. characteristic zero; $\gamma = -1/2$; cubic splits over stated field if rational branches requested

Map. $\alpha = \text{product}/4$, $\beta = -\text{pair_sum}/2$, $\gamma = -1/2$

Return. recover unordered roots; retain selected factor and companion η for ordered inverse

Fibre. 6-to-1 geometrically

Retained information. Rational coefficients do not imply rational roots.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/cayley_fable_bridge/note.tex:131, eq:esbase.

M26 – Fable chart to normalized factor pair. Source. (u,v,w) in affine3

Target. (L,Q) , $\text{Res}(L,Q)=1$ and U2V coefficient1

Domain. characteristic zero; factor coefficients retained

Map. $L = uU + (1+uv)V$; Q as explicit polynomial chart

Return. $u = L_U$; $v = 2L_V Q_{UV} - L_U Q_{VV}$; $w = -4Q_{UV}^2 - 2Q_{UU} Q_{VV} - 12L_V Q_{UV}^2 - 6L_V Q_{UU} Q_{VV} + 9Q_{VV}^2$

Fibre. polynomial isomorphism to normalized factor variety

Retained information. Fable u is not ES divisor u .

Status. retained

Evidence. polynomial source replay plus432 factor-chart inversions

Locators. inputs/cayley_fable_bridge/note.tex:50, eq:factor.

M27 – Factor multiplication. Source. normalized (L, Q)

Target. binary cubic coefficients

Domain. $\text{Res}(L, Q)=1$; generic separable cubic has three simple projective roots

Map. $(L, Q) \rightarrow LQ$

Return. choose one simple root; finite root r gives $u=1/d, v=-r-d, w=5d^2+3rd-\gamma d^3$

Fibre. 3 geometric points off discriminant, 1 over double+simple, 0 at triple root in this chart

Retained information. Constant nonzero Jacobian is local: it does not supply a global single-valued inverse.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/cayley_fable_bridge/note.tex:56, thm:fibres; inputs/cayley_fable_bridge/note.tex:58, eq:inverse.

M28 – Quarter cubic to arbitrary ES reciprocal cubic. Source. three base normalized factors plus ordered roots

Target. three transported normalized factors

Domain. roots distinct; matrix M determinant $D=4(t_2-t_3)(t_1-t_2)(t_1-t_3) \neq 0$

Map. $L=(8/D)L_{\text{star}}$ composed adj M ; $Q=(1/64)Q_{\text{star}}$ composed adj M

Return. undo both scalars and adj M

Fibre. singleton with M and selected factor

Retained information. M and two different scale factors must remain; resultant transforms by D^2 .

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/cayley_fable_bridge/note.tex:171, thm:mobius; inputs/cayley_fable_bridge/note.tex:193, eq:transportpair.

M29 – Tetrahedral pencil states. Source. 12 oriented pairs (H, K)

Target. 3 normalized factors of fixed C_{star}

Domain. H half-sign tetrahedron; simultaneous A_4 action on numerator and denominator

Map. pick component L and complementary Q , then normalize resultant

Return. retain tetrahedral numerator frame to choose the preimage

Fibre. 4-to-1; 24 states if orientation reversals included

Retained information. Four frames above each factor are not four geometric preimages of the cubic.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/tetrahedral_markings/note.tex:130, thm:twelve.

M30 – Rational projective reduction. Source. marked rational projective coordinates

Target. $P_1(F_{23})$

Domain. choose primitive integral pair, not both divisible23; denominators treated projectively

Map. $[a:b] \rightarrow [a \bmod 23 : b \bmod 23]$

Return. no arithmetic inverse unless original rational lift retained

Fibre. infinite fibres

Retained information. The three named marks reduce to infinity,12,11. Do not reduce arbitrary complex Mellin values.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/mellin_leech_research/note.tex:62, eq:projective; inputs/mellin_leech_research/note.tex:67, eq:cycle.

M31 – Abstract flag to projective Leech coordinate. Source. oriented A_4 flag plus orbit bit

Target. 24 named coordinates

Domain. explicit generator matching C, J and base coordinate; order fixed

Map. branch_table(orbit,u,j)

Return. read table inverse

Fibre. singleton finite label map

Retained information. This only transports labels. p , denominators, scales and factors are separate retained fields.

Status. retained

Evidence. original fullfinitechecks and newgenerator-equivariancecomputation

Locators. inputs/mellin_leech_research/note.tex:119, eq:labels.

M32 – Minimal-vector congruence marking. Source. 24 named coordinates

Target. 24 classes in $\Lambda/12\Lambda$

Domain. $v_i = (\text{ones} - 4e_i)/\sqrt{8}$; minimum norm 4

Map. $i \rightarrow [v_i]$

Return. read one of the 24 marked classes

Fibre. singleton on these 24 marks only

Retained information. The full lattice quotient has kernel 12Λ and size 12^{24} , not 24.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/mellin_leech_research/note.tex:175, thm:mod12.

M33 – Three integral lattice traces. Source. Λ_G

Target. Z_3

Domain. three specified octads; no extra congruence condition

Map. $P_j(x/\sqrt{8}) = \sum B_j x_i/4$

Return. $\sigma(t) = \sum t_i b_i$ plus arbitrary k in K_{21}

Fibre. affine rank 21 lattice, $\det 64$

Retained information. P alone loses the kernel; the full (t, k) representation is bijective.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/leech_trace_inverse/note.tex:92, thm:inverse.

M34 – Shortest integral trace lift. Source. t in Z_3

Target. chosen minimum vector $\lambda_{\text{star}}(t)$

Domain. $r = t \bmod 4$, reduced a_i in $\{-1, 0, 1, 2\}$; 64 classes

Map. $\nu_r + \sum ((t_i - a_i)/4) w_i$

Return. $P(\lambda_{\text{star}}) = t$; inverse fibre has all minimizers or full K_{21} as selected target dictates

Fibre. section, not a bijection onto Λ

Retained information. Tie-break choice need not be symmetry-equivariant. New8-seedproof replaces fullshell enumeration for attainment.

Status. retained with shorter standalone proof and strengthened output validation

Evidence. old full196560 replay; new8-seed/64classproof; mutation tests

Locators. inputs/leech_trace_inverse/note.tex:126, thm:shortest.

M35 – Tetrahedral kernel cocycle. Source. (t, k) in Z_3 direct-sum K_{21}

Target. same full-coordinate object

Domain. g in A_4 , ρ_g its induced phase action

Map. $(\rho_g t, gk + g \sigma(t) - \sigma(\rho_g t))$

Return. apply g inverse and corresponding cocycle

Fibre. singleton

Retained information. No invariant integral section outside $4Z_3$. Dropping the correction changes the action.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/leech_trace_inverse/note.tex:201, eq:cocycle; inputs/leech_trace_inverse/note.tex:180, thm:invariants.

M36 – Four-view integral inverse. Source. four ordered triples of common sum

Target. Λ_G plus rank 15 fibre

Domain. only common sum constraints; original 9-column section fixed

Map. $a = \text{pack first triple plus first 2 of other 3}$; $\lambda = \sigma_9(a) + k_9$

Return. all four $P(T^n \lambda)$, residual $\lambda - \sigma_9(a)$

Fibre. affine rank 15, $\det 5292$

Retained information. All 12 T-readings determined; J does NOT descend to this quotient.

Status. retained; J -composition boundary added

Evidence. fullreplay and newexplicitK9/Jcounterexample; repairedCLIchecksallviews

Locators. inputs/leech_trace_inverse/note.tex:222, thm:nine; inputs/leech_trace_inverse/note.tex:217, eq:Tphase.

M37 – Golay to Wilson model. Source. Lambda_G raw24

Target. Lambda_W in O3

Domain. explicit aligned pi,signs; z_j=half rawblock; Wilson normhalf

Map. y_i=epsilon_i x_pi(i)

Return. x_pi(i)=epsilon_i y_i

Fibre. singleton entirelattice

Retained information. Previously absent coordinate arrow now supplied. Existing Wilson-MOG isometry is primary antecedent.

Status. new coordinate completion

Evidence. 37 generating vectors, signed metric identity, unimodularity; allbasis inverse checks

Locators. inputs/cube_unification/note.tex:554, eq:wilson; inputs/leech_trace_inverse/note.tex:36, eq:leech.

M38 – Off-diagonal Albert embedding. Source. Wilson Leech O3

Target. H3(O_R) with zero diagonal

Domain. three octonionic entries and conjugates in fixed positions

Map. $\text{iota}(z) = [[0, z_2, \bar{z}_1], [\bar{z}_2, 0, z_0], [z_1, \bar{z}_0, 0]]$

Return. read ordered off-diagonal entries

Fibre. singleton on image

Retained information. Preserves quadratic norm with factor4; cubic nu is not trace product.

Status. retained with concrete Golay source and norm correction

Evidence. written source and bounded original replay

Locators. inputs/cube_unification/note.tex:569, eq:leechinvariants.

M39 – Faithful combined trace-Albert graph. Source. lambda in Lambda_G

Target. $\text{Xhat} = \text{diag}(P \text{ lambda}) + \text{iota}(Pi \text{ lambda})$

Domain. full graph membership: compact octonions, lattice and 3diagonalconstraints

Map. augment the aligned off-diagonal embedding by its three traces

Return. read all z, invert signed permutation, recompute traces

Fibre. singleton

Retained information. $4N - pS = F_p(t) + \sum(p - 4t_i)N(z_i) + 4nu$. Every correction retained.

Status. new corrected interface

Evidence. written cubic formulas, exactQ(i) checks, full2600-node determining check

Locators. inputs/leech_trace_inverse/note.tex:262, eq:defect; inputs/cube_unification/note.tex:333, eq:titsnorm.

M40 – Albert coordinates to 27 line variables. Source. three 3x3 typed blocks

Target. 27 independent labeled scalar variables

Domain. specific ordered bijection in source; char0

Map. dictionary gives 45 signed norm monomials matching 45 tritangent triples

Return. read matrix entries by labels

Fibre. singleton coordinate map

Retained information. Not an isomorphism of the cubic surface with 27-dimensional affine space. Weyl lifts may retain signs after permutations return.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/cayley_fable_bridge/note.tex:261, eq:blocks; inputs/cayley_fable_bridge/note.tex:272, thm:dictionary.

M41 – Full orbit trace completion. Source. lambda in Lambda_G

Target. a_B for all 759 octads

Domain. explicit linear image equations and exact mod 4 syndrome

Map. $a_B = \sum_B x/4$

Return. $x_i = (23 S_i - 7S)/1012$; lattice image iff $(11S - 23S_0)/506 = 0 \pmod{4}$

Fibre. singleton

Retained information. Smallest real stable span under $\langle J, T \rangle$ has rank 24; no 9-dimensional full group quotient.

Status. new integral reconstruction theorem

Evidence. 759 orbit, Gram 576, 24 minor $\det 2^{14}$, all integer dual quotient proof

Locators. inputs/leech_trace_inverse/note.tex:222, thm:nine; inputs/mellin_leech_research/note.tex:88, thm:groups.

M42 – Vector theta Fourier return. Source. $\Lambda/12\Lambda$ theta packet

Target. same packet after τ or t inversion

Domain. fullfinite module size 12^{24} , bilinear pairing mod 12; preserve Jacobi variable z for classidentification

Map. $F_{ab} = 12^{-12} \exp(-2\pi i(a,b)/12)$

Return. F^2 sends a to $-a$

Fibre. invertible on fullpacket; selected 24 marks not closed

Retained information. Setting $z=0$ can identify isometric cosets; selecting 24 classes loses Fourier support.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/mellin_leech_research/note.tex:384, eq:vectorTheta.

M43 – Mellin transform of literal source kernel. Source. $E h$ with $h = (\pi^2 x^4 - 3\pi x^2/2)e^{-\pi x^2}$

Target. $\xi(s+1/2)/4$

Domain. source convention $u^{-(s-1)}$; finite kernels and centered strip as stated

Map. $M(Eh) = \xi/4$; normalized source is $4Eh$

Return. Mellin inversion requires analytic domain/growth and all transform data; no pointwise inverse claimed here

Fibre. not a finite projective rootmap

Retained information. Completion marks $\pm 1/2$ are not ξ zeros; infinity is chart mark not integral substitution.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/mellin_leech_research/note.tex:42, prop:normalization.

M44 – Residue-valued arithmetic kernel. Source. all positive integers n and residue operators

Target. matrix Mellin Dirichlet packet

Domain. $R_n = 0$ for $\gcd(n, 12) > 1$; actual n retained

Map. $B(u) = 4\sqrt{u} \sum R_n h(nu)$

Return. retain all 4 character projections and phases; restore 2, 3 factors by $full^2 a^3 b$ sum

Fibre. discarding characters loses components

Retained information. Finite projective labels are not arithmetic values; imprimitive factors and conductors retained.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/mellin_leech_research/note.tex:231, thm:dirichlet; inputs/mellin_leech_research/note.tex:250, eq:restore.

M45 – Leech theta centered kernel. Source. selfdual positive definite lattice rank $2kappa$

Target. entire even Mellin function

Domain. $\Psi = u^{1/2} \Theta(u^{1/kappa})$; zero-mode killed; Leech $kappa 12$

Map. $K = 1/2(D^2 - 1/4)\Psi$; $w = kappa(s+1/2)$

Return. retain full theta coefficients/analytic data, not just end point values

Fibre. spectral scalar observations can lose vector coset labels

Retained information. Leech transform has $\zeta(w)\zeta(w-11)-L(\Delta, w)$, not Riemann ξ .

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/mellin_leech_research/note.tex:316, thm:theta; inputs/mellin_leech_research/note.tex:361, eq:LeechM.

M46 – Finite kernel approximation. Source. h_λ compact approximation with explicit C

Target. Mellin B_λ converging on closed substrips

Domain. $\epsilon > 0$, stated error; retain tail and end point estimates

Map. source finite kernel construction

Return. no spectral inverse to Weillowesteigenfunction proved

Fibre. N/A analytic estimate

Retained information. Kernel approximation is not eigenfunction comparison or RH proof.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/mellin_leech_research/note.tex:257, thm:error.

M47 – Terzo exponential evaluation. Source. free exponential-ring quotient A

Target. complexnumbers

Domain. specified relations; injection requiresSchanuel, notusedunconditionally

Map. evaluate(X,Y)=(pi,i)

Return. kernelJ; cube evaluation kernelJ²⁷

Fibre. unknown unconditionalkernel

Retained information. Period subgroupZ varpi injects byordinaryevaluation, but fullringfaithfulnessconditional.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/cube_unification/note.tex:882, eq:periodmap.

M48 – Arithmetic cubic selection. Source. complete positiveintegraltracefibres

Target. ESsolution locus

Domain. p prescribed prime; Fp=0 actualequation; allmarks inherited

Map. select t>0 with 4t0t1t2=p pair_sum

Return. selector reconstructionwhenboundedmarkexists; no universal producerproved

Fibre. each t hasfull latticekernel-fibre

Retained information. Surjectivity ofP doesnotestablish existenceoft withFp=0.

Status. explicit unresolved existence arrow

Evidence. written source and bounded original replay

Locators. inputs/leech_trace_inverse/note.tex:262, eq:defect; inputs/cube_unification/note.tex:804, prop:prime-return.

M49 – Fixedmodulus Selberg conclusion. Source. fixedrayfamily andactualprimefactoroccurrences

Target. explicit exceptional-count theorem

Domain. allowedprime restriction applies to every prime/divisor sum; square majorant before CRT; fixedfamily constants

Map. count exceptions to existence of bounded residual

Return. notapointwise inverse; no movingmodulusPNT silentlyused

Fibre. N/A countingtheorem

Retained information. No finitecomputation or densityone leap to universal selector.

Status. retained

Evidence. written source and bounded original replay

Locators. inputs/es_pointwise_tranche/tranche.tex:556, app:sieve; inputs/erdos_straus_research/paper.tex:456, thm:small.

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Additional exact antecedents. Yamamoto (1965), Lemma 2, equation (4), p. 38; Elsholtz–Tao (2013), Section 2; the precise references and original evidence remain with their pinned source modules. Tao’s 21 July 2026 exposition supplies the Fable factor construction: <https://terrytao.wordpress.com/2026/07/21/a-digestion-of-the-jacobian-conjecture-counterexample/>. Baez–Pumplün (2026), Sections 5–6, Theorem 11 and Appendix A, provide the classical three-block and Albert interface used in the code: <https://arxiv.org/html/2609.07538v1>. Connes–Consani–Moscovici, *Zeta spectral triples*, version 1, Section 7: <https://arxiv.org/html/2511.22755v1>. None of these references is substituted for an unperformed complete proof audit.